AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2017

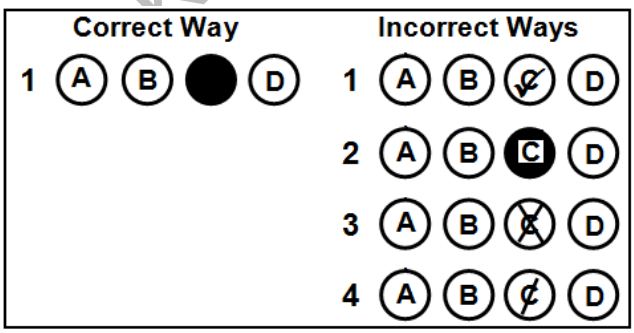
Chemistry Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. What will be the decomposition products of $PbCl_4$ at room temperature?

   A. $Pb + Cl_4$
   B. $Pb + 2Cl_2$
   C. $PbCl_2 + 2Cl^-$
   D. $PbCl_2 + Cl_2$

2. The most stable oxidation state of carbon is

   A. + 4
   B. + 2
   C. – 2
   D. – 4

3. The tetrahalide which has the highest tendency to hydrolyse is

   A. CCl_4
   B. $SiCl_4$
   C. $GeCl_4$
   D. $SnCl_4$

4. All of the following are characteristics of an alloy EXCEPT that it

   A. is corrosive.
   B. is strong and flexible.
   C. has high melting point.
   D. is relatively cheaper than metals.

5. The systematic name of the given transition metal complex is

   $[Co(NO_2)_3(NH_3)_3]$

   A. Trinitro triammine cobalt (III)
   B. Triammine trinitro cobalt (III)
   C. Trinitro triammine cobalt (VI)
   D. Triammine trinitro cobalt (VI)

6. The correct order of reactivity of alkyl halides is

   A. R–I > R–Br > R–Cl > R–F
   B. R–F > R–Cl > R–Br > R–I
   C. R–Cl > R–Br > R–I > R–F
   D. R–Br > R–Cl > R–F > R–I

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7. The product **X** in the given reaction is

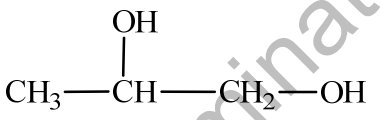


A. $CH_3–CO–CH_3$
B. $CH_3–CH_2–CHO$
C. $CH_3–CH_2–COOH$
D. $CH_3–CH_2–CH_2–OH$

8. Methylamine is more basic than ammonia, because the

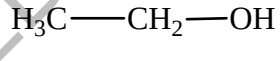
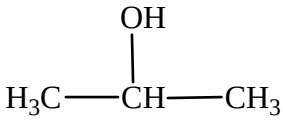
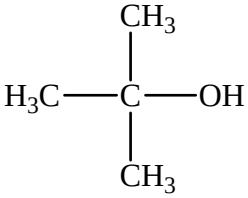
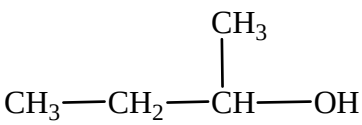
A. methylammonium ion is more stabilised.
B. methylammonium ion is a stronger conjugate base than ammonium ion.
C. sp^3 orbital of carbon in methylamine pulls the electron pair towards itself.
D. electron pair in methylamine is strongly attracted by s-orbital of hydrogen atom.

9. The oxidative cleavage of the given diol produces



A. 2 molecules of ketone.
B. 2 molecules of aldehyde.
C. 1 molecule of ketone and 1 molecule of ester.
D. 1 molecule of aldehyde and 1 molecule of ester.

10. An alcohol which reacts with concentrated HCl in anhydrous $ZnCl_2$ and forms an oily layer immediately is

A	
B	
C	
D	

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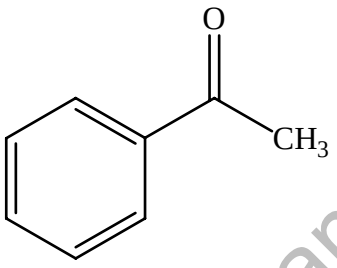
11. The correct order of acidic strength of alcohols, phenols and carboxylic acids is

    A. carboxylic acids > phenols > alcohols.
    B. alcohols > phenols > carboxylic acids.
    C. phenols > carboxylic acids > alcohols.
    D. carboxylic acids > alcohols > phenols.

12. The intermediate product which is formed during the chemical reaction of ether with hydrogen iodide is

    A. carbanion.
    B. oxyanion.
    C. carbocation.
    D. oxonium ion.

13. The CORRECT number of sigma bonds in the given organic compound is



    A. 13
    B. 14
    C. 17
    D. 21

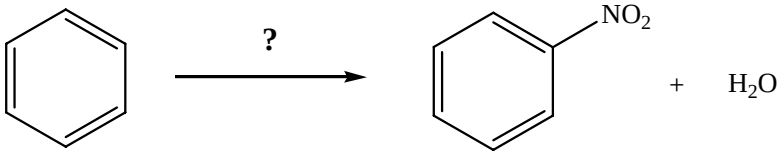
14. The given reaction is an example of



    A. halogenation.
    B. halohydration.
    C. hydroxylation.
    D. hydrohalogenation.

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15. The reagent(s) required for the given chemical reaction is/ are



A. H_2SO_4
B. $NaNO_2$
C. $NaNO_2$ and H_2SO_4
D. HNO_3 and H_2SO_4

16. Oximes are formed when aldehydes or ketones reacts with

A. ammonia.
B. hydroxylamine.
C. phenylhydrazine.
D. 2,4-dinitrophenylhydrazine.

17. An aldehyde reacts with a hydrazine to form a/ an

A. alcohol.
B. osazone.
C. hydrazone.
D. phenylhyrdazone.

18. Which of the following carboxylic acids is the most acidic?

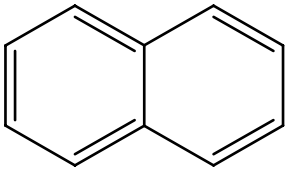
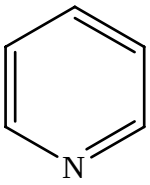
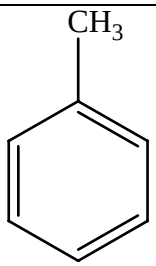
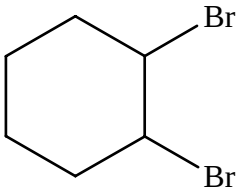
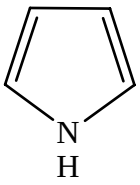
A. $CH_3–COOH$
B. $F–CH_2–COOH$
C. $Cl–CH_2–COOH$
D. $CH_3–CH_2–COOH$

19. With an increase in the molecular mass of carboxylic acid, its melting point

A. increases regularly.
B. decreases regularly.
C. increases irregularly.
D. decreases irregularly.

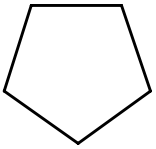
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20. Which of the following is/ are heterocyclic compound(s)?

				
I	II	III	IV	V

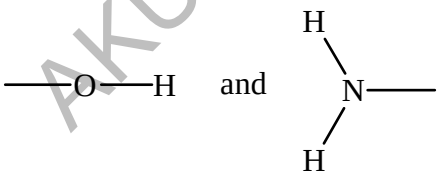
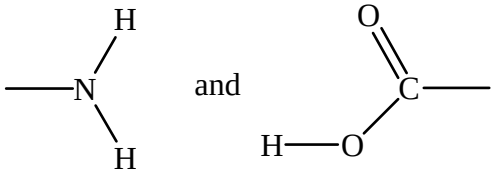
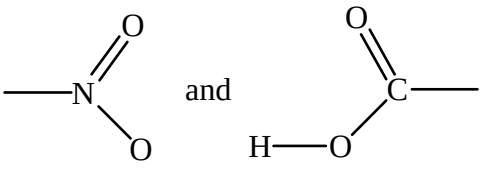
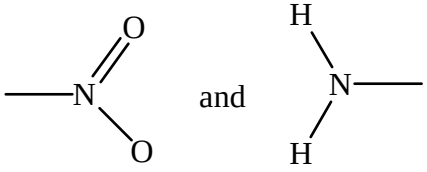
A. I only
B. IV only
C. II and V
D. III and IV

21. The given organic compound can be classified as a/ an



A. aliphatic compound.
B. alicyclic compound.
C. aromatic compound.
D. heterocyclic compound.

22. If reacted, which of the following functional groups will give rise to an amide (peptide) linkage?

	
A	B
	
C	D

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23. Milk proteins get denatured when lemon juice is added to milk. The process of denaturation affects all EXCEPT

A. tertiary structure of proteins.
B. primary structure of proteins.
C. biological activity of proteins.
D. secondary structure of proteins.

24. Nylon-6,6 is a polymer which is formed by the combination of

I. adipic acid
II. vinyl chloride
III. hexamethylene diammine

A. I only
B. I and II
C. I and III
D. II and III

25. Which of the following is FALSE about condensation polymerisation?

A. Small molecules are eliminated during the reaction.
B. Bifunctional monomers react to form long chain polymers.
C. Unsaturated monomers react together to form long chain polymers.
D. Condensation polymers form slowly as compared to addition polymers.

26. Urban smog results in the formation of peroxyacetyl nitrate which causes damage to

A. plant life.
B. rubber tyres.
C. kidney tissues.
D. the nervous system.

27. The acid which has the least harmful effect on animals and plants is

A. HCl
B. HNO_3
C. H_2CO_3
D. H_2SO_4

28. Which of the following compounds can easily be identified with the help of UV-visible spectroscopy?

A. $C(CH_3)_4$
B. $CH_3–(CH_2)_3–CH_3$
C. $(CH_3)_3C–CH_2–CH_3$
D. $CH_3–CH=CH–C(CH_3)_3$

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29. The electromagnetic radiation which has the shortest wavelength in electromagnetic spectrum is

    A. X-ray.
    B. infrared.
    C. microwave.
    D. gamma ray.

30. One of the disadvantage of using emission spectroscopy is that it

    A. gives low precision value.
    B. determines metalloids only.
    C. is an extremely sensitive technique.
    D. requires minimum amount of sample.

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